

Highlights

Dual-Field Electricity Price Forecasting with Predispatch

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- A trend–event dual-field model is adapted from reconstruction to price forecasting.
- Static-fit fields worsen MAE but improve CRPS relative to the same linear base.
- Quarterly refits and predispatch yield pooled gains, with regional differences.
- Negative-price scores improve over all five main-table comparators.
- A validation-selected lower bound trades tail accuracy for lower overall MAE.

Dual-Field Electricity Price Forecasting with Predispach

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Abstract

Wholesale electricity prices combine a changing level with scarcity spikes and negative-price troughs. We adapt the trend–event structure of DualTimesField to rolling 24-hour forecasting using detected historical events, a linear base, and point-in-time market-operator forecasts. Experiments cover three regions of the Australian National Electricity Market, with 17,521 hourly origins per region in 2023–2024. On a single fit to 2015–2021, the fields worsen mean absolute error (MAE) relative to the same linear base, although they improve its approximate continuous ranked probability score (CRPS). With quarterly refits and AEMO predispach inputs, the fields improve both metrics when pooled across regions; point-forecast gains are not uniform across regions. A shared per-hour calibrator and a validation-selected lower bound further reduce MAE. The final model achieves a three-region mean MAE of 36.6 AUD/MWh, below PatchTST, DLinear and a linear baseline (pooled Diebold–Mariano $p < 0.01$), but above XGBoost at 35.5. Its negative-price CRPS is lower than all five main-table comparators ($p < 0.001$). The lower bound trades a small deterioration in negative-price scores for better overall MAE. These results identify the conditions and limits of the decomposition’s forecasting value.

Keywords: Electricity price forecasting, Probabilistic forecasting, Time-series decomposition, Negative prices, National Electricity Market, Predispach

1. Introduction

Short-term electricity price forecasts drive bidding, storage dispatch, demand response and hedging decisions [1, 2]. In markets with a fast-growing

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4 share of wind and solar generation, these forecasts have become harder. Prices
5 still follow a daily and seasonal level, but they are increasingly punctuated
6 by events: scarcity spikes when supply margins tighten and negative-price
7 troughs when renewable output exceeds demand. In the Australian National
8 Electricity Market (NEM), the share of hours with negative prices in Queens-
9 land rose from 1.6% in 2015–2021 to 13.2% in 2023–2024 (Section 3). A
10 forecaster that treats these events as noise around a level misses exactly the
11 hours in which storage and flexible demand earn or lose most.

12 Separating a series into a continuous trend and discrete events is therefore
13 an attractive structural prior. Classical decompositions and their neural
14 successors [3, 4, 5] split trend from seasonality; DualTimesField [6] goes
15 further and represents a series as a continuous trend field (CTF), an implicit
16 neural representation in continuous time [7, 8], plus a discrete event field
17 (DGF) of localized atoms. It was designed for reconstruction and interpolation.
18 Whether such a representation helps forecasting is not obvious: describing
19 the past more faithfully does not by itself add information about the future,
20 and a richer description of a past regime can overfit it.

21 This paper studies that question for rolling 24-hour price forecasting in
22 three NEM regions, under a same-data protocol in which every compared
23 model receives the same underlying inputs, the same training periods and the
24 same evaluation. Our findings are sequential, and the paper follows them:

- 25 1. *Transplanted, the dual field does not separate events.* The CTF absorbs
26 spikes and the DGF collapses. Detecting events explicitly, as the
27 largest departures from the window median, restores the separation
28 (Appendix A).
- 29 2. *Separation alone does not improve point accuracy.* On a single fit to
30 2015–2021, a linear model on the same inputs has lower MAE than
31 the dual-field model. Adding the fields to the same linear base raises
32 MAE but lowers CRPS (Section 6.2). Lower training loss and higher
33 validation MAE are consistent with overfitting to the training-period
34 price regime.
- 35 3. *Recalibration and market-operator information yield pooled gains.* Moti-
36 vated by the frequent recalibration used in electricity price forecasting
37 [9], we refit every model before each test quarter. AEMO’s predispach
38 combines current offers, constraints and demand forecasts, and lowers
39 every model’s three-region mean MAE. With these inputs and refits,
40 the dual fields improve pooled MAE and CRPS over the same linear

base; raw-price MAE nevertheless worsens in QLD1.

4. *Diagnosis-driven corrections.* A per-region analysis of the errors that remain against the strongest competitor locates them in negative-price hours, where linear heads over-extrapolate in the transformed target space. A shared per-hour calibrator and a price floor chosen on validation data address this and are each significant.

The final model has lower MAE than PatchTST [10], DLinear [5] and a linear model, and lower negative-price CRPS and lower-tail pinball loss than the five main-table comparators. XGBoost [11] with the same inputs and refits remains more accurate on overall MAE and CRPS. The final model is not the best of all development variants on negative-price scores: omitting its lower bound improves those scores but worsens overall MAE. Beyond the model, the study shows why a structured forecaster should be evaluated against strong same-input baselines and under its intended recalibration regime.

The rest of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the data and the forecasting task, Section 4 the model, and Section 5 the experimental protocol. Section 6 reports the results, Section 7 discusses them, and Section 8 concludes.

2. Related work

Electricity price forecasting. Reviews by Weron [1] and Nowotarski and Weron [12] survey point and probabilistic electricity price forecasting. Lago et al. [9] established an open benchmark and best practices, among them daily recalibration on a rolling window, variance-stabilizing transformations [13], and comparison against well-specified linear (LEAR) and neural baselines with Diebold–Mariano tests [14]. High-dimensional linear models [15] and distributional neural networks [16] are strong references. We follow these practices: an asinh transformation, rolling recalibration, same-input baselines, and significance tests.

Deep time-series forecasting and decomposition. Transformer forecasters such as Informer [17], Autoformer [3], PatchTST [10] and iTransformer [18], recurrent models [19, 20] and DeepAR [21] have been widely applied. Zeng et al. [5] showed that a decomposed linear model (DLinear) is competitive with many of them. N-BEATS [4] and Autoformer use trend and seasonal decompositions inside the network; hybrid models combine a linear component

75 with a nonlinear residual [22]. DualTimesField [6] represents trend and events
 76 as continuous-time fields for reconstruction. Our model adds dual fields as a
 77 residual to a linear base, and asks when this structure pays off in forecasting.

78 *Exogenous information and strong tabular baselines.* Fundamental forecasts
 79 of load, renewable output and capacity are established inputs in electricity
 80 price forecasting [1, 9]. Recent work adds text: Chen et al. [23] introduce
 81 RE-Price, which combines structured news extraction, cross-modal attention
 82 and a QLoRA-adapted Qwen3-4B backbone. Monte Carlo dropout under
 83 standard and tail-aware prompts generates scenarios, which are smoothed
 84 with adaptive Gamma kernels. Their study uses the same three NEM re-
 85 gions and the 2015–2024 period, with a chronological 70/10/20 split and
 86 72-hour inputs for 24-hour forecasts. However, its inputs include WattClarity
 87 news and temperature-forecast proxies constructed from noisy observations,
 88 whereas our main setting uses numerical market-operator forecasts with quar-
 89 terly refits. The information sets and evaluation protocols therefore differ;
 90 RE-Price is related work, not a reproduced same-data competitor, and its pub-
 91 lished errors are not directly comparable with ours. The strong performance
 92 of gradient-boosted trees on tabular benchmarks [24] motivates including
 93 XGBoost alongside neural baselines.

94 **3. Data and forecasting task**

95 *Market and prices.* We use hourly regional reference prices (RRP) and op-
 96 erational demand for New South Wales (NSW1), Queensland (QLD1) and
 97 Tasmania (TAS1) from AEMO’s aggregated price and demand data [25],
 98 1 January 2015 to 31 December 2024 in market time (AEST). Half-hourly
 99 records (to September 2021) and five-minute records (from October 2021,
 100 when five-minute settlement began) are averaged to hours. Table 1 summa-
 101 rizes the series. The test period, 2023–2024, differs markedly from 2015–2021:
 102 negative prices are about three to seventeen times more frequent (2.9 times
 103 in TAS1, 8.1 in QLD1 and 16.8 in NSW1), spikes above 300 AUD/MWh are
 104 more frequent in NSW1 and QLD1, and 2022, the energy-crisis year that
 105 separates them, has a much higher level.

106 *Forecasting task.* Let p_t denote the price for the delivery hour starting at
 107 t . At origin t , the target is $(p_t, \dots, p_{t+23})$, using observations from hours
 108 $t - 72, \dots, t - 1$ and exogenous information published by t . The observation
 109 for hour $t - 1$ becomes available at t ; no target-hour observation enters the

Table 1: Hourly regional reference prices (AUD/MWh) by period. Neg.: share of hours below zero; > 300: share of hours above 300 AUD/MWh.

Region	Period	Mean	Median	Std. dev.	Neg.	> 300
NSW1	2015–2021	70.5	58.0	164.9	0.37%	0.25%
	2022	182.7	135.4	215.3	2.61%	14.79%
	2023–2024	113.5	82.7	408.8	6.20%	1.75%
QLD1	2015–2021	70.9	55.2	202.5	1.63%	0.68%
	2022	205.1	139.9	456.2	4.05%	15.73%
	2023–2024	101.1	81.3	289.2	13.21%	1.85%
TAS1	2015–2021	69.4	57.6	68.7	2.53%	0.43%
	2022	147.0	104.9	285.3	1.82%	8.57%
	2023–2024	75.8	60.3	168.2	7.36%	0.50%

history. We index the 24 forecast outputs by $h = 0, \dots, 23$. Origins advance hourly, giving 17,521 complete test windows per region. This is a rolling 24-hour task, distinct from a fixed-origin day-ahead auction forecast.

Known inputs. Four groups of forward-looking information enter every model, each as published at the origin:

- *Calendar:* hour of day, day of week and day of year (sine and cosine), month, weekend flag and a standardized year.
- *PD PASA:* AEMO’s pre-dispatch projected assessment of system adequacy [26], from which we take maximum spare capacity and net load (50%-probability demand minus semi-scheduled wind and solar availability) for each forecast hour, from the latest run published by the origin, with the short-term PASA run filling hours beyond its horizon.
- *Gas:* the logarithm of the seven-day trailing mean of the Victorian declared wholesale gas market price [27], using only gas days ended before the origin.
- *Predispatch:* AEMO’s predispatch re-solves the market dispatch every 30 minutes with current offers, constraints and demand forecasts. Its horizon extends to the end of the trading day for which bid prices have closed, and expands to include the next trading day at 12:30 AEST

[28]. We obtain regional price and quantity forecasts from the MMSDM archive [26]. For each origin we take the latest run whose rows were all published by then and keep the forecast price, total demand, available generation and net interchange for each hour. Runs published between about 04:00 and 12:30 end before the 24th forecast hour; uncovered hours repeat the last covered hour and are flagged (93% of hours are covered). The archive has no run history for October 2022, whose origins are marked uncovered and filled with training-period medians.

Used directly as a forecast, the predispatch price has a test MAE of 147.5, 153.3 and 69.4 AUD/MWh in NSW1, QLD1 and TAS1, inflated by spikes that do not materialize, but a median absolute error of only 17–20 AUD/MWh: it is informative but must be calibrated.

Price treatments. We evaluate on raw prices and, as a robustness check that limits the weight of a few extreme hours in absolute-error metrics, on prices capped at 650 AUD/MWh, with the cap applied to the price series used as target and as history, in training and in scoring.

4. Method

Fig. 1 gives an overview. The forecast is the sum of a linear base on all inputs, a dual-field residual built from a trend–event decomposition of the history, and a calibration term, followed by a price floor. Every component is refit before each test quarter.

4.1. Target transformation

Prices are mapped by the area hyperbolic sine with median and median-absolute-deviation scaling [13],

$$y = \operatorname{asinh}\left(\frac{p - \mu}{\sigma}\right), \quad \mu = \operatorname{med}(p^{\text{train}}), \quad \sigma = \frac{\operatorname{med}|p^{\text{train}} - \mu|}{0.6745}, \quad (1)$$

and standardized with training statistics. The point forecast is produced in this space and mapped back by the inverse transform. The quantile forecasts are produced directly in standardized price space, which keeps their spread in the units on which they are scored.

4.2. Dual-field decomposition of the history

Let $x \in \mathbb{R}^{72 \times 2}$ be the standardized history of transformed price and demand, indexed by $\tau = 1, \dots, 72$.

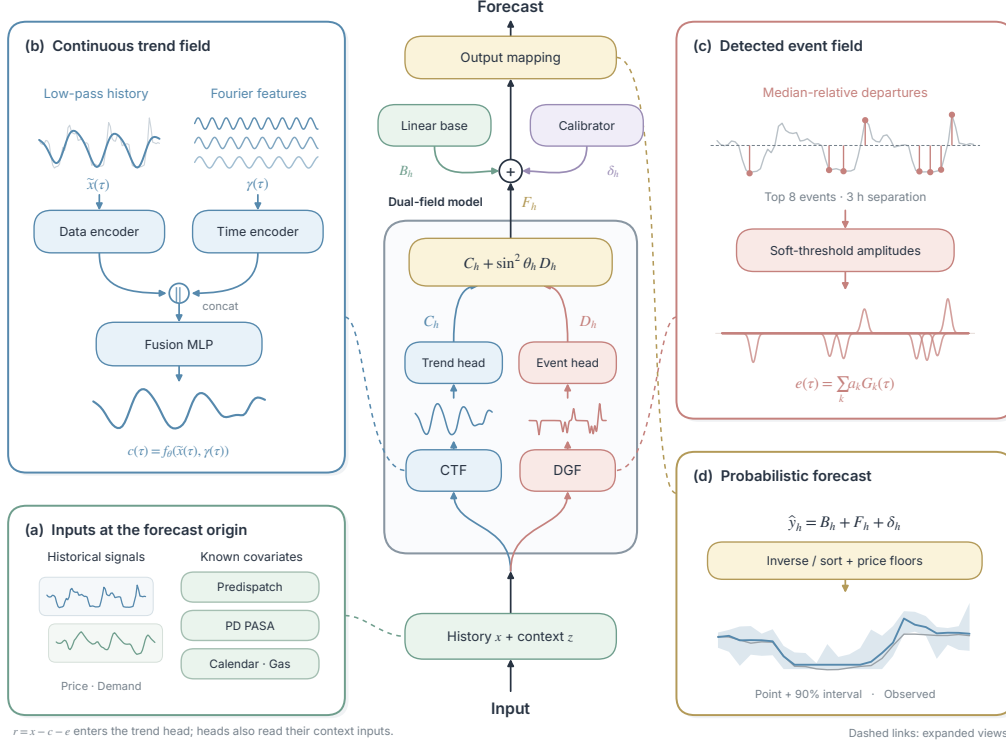


Figure 1: Central forecasting architecture and expanded mechanisms. Solid arrows show computation; dashed links identify expanded views. (a) The inputs comprise 72 hours of price and demand history and covariates known at the forecast origin. (b) The CTF encodes low-pass full history and Fourier time features in parallel, concatenates their representations and produces the trend. (c) The DGF detects price departures from the window median and sums soft-thresholded Gaussian atoms. The central heads form the gated field residual F_h , which is added to the linear base B_h and shared calibration δ_h . Head-specific context inputs, the remainder $r = x - c - e$, and quantile-specific gates are detailed in Section 4. (d) Point predictions are inverse-transformed; price-space quantiles are sorted, and separate validation-selected lower bounds apply. Signal thumbnails use the QLD1 case of 1 October 2023 and the seed-2026 refit. The forecast shows 24 hours, its 90% interval and observations in gray; the Fourier inset shows three learned bases.

160 *Event field (DGF)*. Events are detected in the price channel $x^{(1)}$ relative
 161 to a robust level. With departures $d_\tau = x_\tau^{(1)} - \text{med}_\tau x_\tau^{(1)}$, we select $K = 8$
 162 positions $c_1, \dots, c_K$ greedily by largest $|d_\tau|$, suppressing positions closer than
 163 three hours to an already selected one. Each event keeps its departure,
 164 soft-thresholded by a learned level $\kappa > 0$, and becomes a Gaussian bump of
 165 learned width w :

$$e(\tau) = \sum_{k=1}^K a_k \exp\left(-\frac{(\tau - c_k)^2}{2w^2}\right), \quad a_k = \text{sign}(d_{c_k}) \max(|d_{c_k}| - \kappa, 0). \quad (2)$$

166 Event positions are determined by the history, while the threshold and
 167 width remain learned parameters ($w = \text{softplus}(w_0) + 0.25$ hours). The
 168 event signal is nonzero only in the price channel; its demand channel is zero.
 169 This constrains the decomposition more directly than globally learned event
 170 positions (Appendix A); it does not impose a fixed event signal throughout
 171 training.

172 *Trend field (CTF)*. The CTF is a Fourier-feature implicit neural representa-
 173 tion [8, 6]. With normalized time $\tau' \in [0, 1]$, learned band-limited frequencies
 174 $b_j = \omega_{\max} \text{sigmoid}(\beta_j)$, $j = 1, \dots, 16$, and a low-pass filtered copy $\tilde{x}_\tau$ of the
 175 history,

$$\gamma(\tau') = \left[\cos 2\pi b_j \tau', \sin 2\pi b_j \tau' \right]_{j=1}^{16}, \quad c(\tau) = f_\theta(\gamma(\tau'), \tilde{x}_\tau), \quad (3)$$

176 where f_θ is a small multilayer perceptron. The CTF receives the low-pass
 177 filtered full history, not an event-subtracted input. The decomposition loss
 178 $\mathcal{L}_{\text{dec}} = \|c + e - x\|_2^2 / |x|$ encourages it to account for the signal not represented
 179 by the events, and $r = x - c - e$ is the remainder. The net-load input to its
 180 forecast head is standardized on training data, with logarithmic compression
 181 beyond the training 0.5% and 99.5% quantiles.

182 4.3. Forecast heads, linear base and fusion

183 For each of the 24 forecast hours h , linear heads map field features to a
 184 point forecast in transformed space and to quantiles at the levels 0.05, 0.10,
 185 0.50, 0.90 and 0.95 in price space:

- 186 • *CTF heads* read the trend c , the remainder r , the calendar, the gas
 187 price and the PD PASA net load: the level and its fundamental drivers.

188 • *DGF heads* read the event field e , the calendar, the PD PASA spare
 189 capacity and the predispatch inputs; the DGF quantile head also reads
 190 the shortfalls of spare capacity below its training 1%, 5% and 10%
 191 quantiles, a piecewise-linear scarcity response.

192 The two are fused by an additive trigonometric gate, $\hat{y}_h^F = \hat{y}_h^C + \sin^2(\theta_h) \hat{y}_h^D$,
 193 with $\theta_h = \frac{\pi}{2} \text{sigmoid}(g_h)$ and g_h linear in the fields, the calendar and the DGF
 194 inputs; each quantile level has its own affine rescaling of g_h , so tail quantiles
 195 can draw on the event field more than the median.

196 The *linear base* $\hat{y}^B$ is one linear map from the standardized, undecomposed
 197 history and all known inputs to the 24 point and quantile outputs. The fused
 198 field forecast is added to it as a residual, so the fields only need to model
 199 what a linear function of the inputs cannot.

200 *Calibrator.* A single perceptron with two 32-unit hidden layers and GELU
 201 activations, shared by all forecast hours, reads hour h 's known inputs (predis-
 202 patch, PD PASA, calendar), a learned four-dimensional horizon embedding
 203 and two transformed-price summaries (last observed value and 24-hour mean),
 204 and outputs additive corrections δ_h to the point and quantile forecasts. Its
 205 output layer is initialized at zero. It lets the model learn a nonlinear response
 206 to predispatch, for example that high predispatch prices are often not realized.
 207 The point forecast before inverse transformation and the lower bound is

$$\hat{y}_h = \hat{y}_h^B + \hat{y}_h^F + \delta_h. \quad (4)$$

208 *Training.* The recorded training and checkpoint-selection loss is

$$\mathcal{L} = \underbrace{\frac{1}{24} \sum_h (\hat{y}_h - y_h)^2}_{\text{point}} + \underbrace{\frac{1}{24.5} \sum_{h,q} \rho_q (\tilde{p}_h - \hat{p}_{h,q})}_{\text{quantiles}} + \mathcal{L}_{\text{dec}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_g \mathcal{L}_{\text{sparse}}, \quad (5)$$

209 where ρ_q is the pinball loss [29], $\tilde{p}$ is the standardized price, and $\mathcal{L}_{\text{smooth}}$
 210 penalizes squared first differences of the CTF. The retained implementation
 211 computes $\mathcal{L}_{\text{sparse}} = \bar{z} + 10 \max(\bar{z} - 0.3, 0)$, where $z_k = \mathbf{1}[a_k \neq 0]$ and $\bar{z}$ is
 212 the batch mean ($\lambda_s = \lambda_g = 10^{-3}$). This hard indicator has no gradient: the
 213 sparsity term contributes to the reported loss and checkpoint selection, but
 214 does not directly regularize parameter updates. Event sparsity instead results
 215 from selecting at most eight positions and soft-thresholding their amplitudes;
 216 the threshold and width receive gradients through the reconstruction and
 217 forecasting terms. We use AdamW with learning rate 3×10^{-4} , weight decay

218 10^{-4} , batch size 128 and 30 epochs, and keep the epoch with the lowest
 219 recorded validation loss. The model has 277,747 parameters. Quantiles are
 220 sorted after field fusion, addition of the base, and calibration to prevent
 221 crossings.

222 4.4. Price floor

223 The fitted linear heads can extrapolate to very low prices after the inverse
 224 asinh transform (Section 6.3). We apply an empirical lower bound, $\hat{p}_h \leftarrow$
 225 $\max(\hat{p}_h, \phi)$; this is a forecast calibration choice, not the NEM’s regulatory
 226 market price floor or a physical bound on negative prices. For each refit
 227 and region, ϕ is chosen on that refit’s validation quarter from $\{\text{none}\} \cup$
 228 $\{\min(Q_\alpha(p^{\text{train}}), 0) : \alpha \in \{0.05, 0.1, 0.2, 0.5, 1\}\%\}$. The training-price sample
 229 here pools the training target windows. Point and quantile lower bounds are
 230 selected separately by mean validation MAE and CRPS over the three seeds,
 231 respectively. No test-quarter prices are used to select the numerical bound.
 232 Its effect on negative-price scores is reported separately from its overall MAE
 233 gain.

234 4.5. Quarterly recalibration

235 Before each of the eight test quarters k (2023Q1 to 2024Q4), every model
 236 is refit on data from January 2015 to three months before the quarter, selects
 237 its checkpoint on those three months, and forecasts the origins of the quarter.
 238 Forecasts of the eight quarters together cover exactly the 17,521 test origins
 239 of each region. Transformations, standardizations and quantile knots are
 240 re-estimated on each refit’s training data.

241 5. Experimental setup

242 *Baselines.* All baselines receive the same history and known inputs, including
 243 predispatch, and are refit on the same quarterly schedule.

- 244 • *XGBoost* [11]: one multi-quantile model per forecast hour on the flat-
 245 tened history, gas price and that hour’s calendar and known inputs
 246 (depth 6, learning rate 0.05, subsampling 0.8, up to 2,000 rounds with
 247 early stopping on validation); the median is the point forecast.
- 248 • *PatchTST* [10] (patch length 16, stride 8, instance normalization, three
 249 layers, $d_{\text{model}} = 128$) and *DLinear* [5] (moving-average decomposition).
 250 Both are channel-independent and would otherwise ignore the known

inputs, so each adds a linear map from the known future inputs and the demand history to its outputs.

- *Linear*: one linear map of the price history plus the same linear map of known inputs, with a mean-squared-error point output and pinball-loss quantiles; in our experiments it is the strongest linear specification and dominates DLinear.
- *Linear base only*: our model without the dual fields, the calibrator and the floor, trained by the same pipeline; it isolates the contribution of the structure.

Neural baselines output the five quantiles of the transformed price and are trained with the pinball loss (Adam, batch size 256, 30 epochs, weight decay 10^{-5} , best validation epoch); their learning rates were chosen from $\{10^{-4}, 5 \cdot 10^{-4}, 10^{-3}\}$ on validation data. Two further neural baselines, iTransformer [18] and Informer [17], were less accurate than PatchTST and DLinear in a static-split screening (Appendix B) and are not refit.

Metrics. Point forecasts are scored by MAE and by window RMSE, the root mean squared error of each 24-hour window averaged over windows. Probabilistic forecasts are scored by the coverage and average interval score (AIS) of the 5–95% interval and by an approximation of the continuous ranked probability score [30], twice the mean pinball loss over the five quantiles (CRPS). MAE is minimized by the median [31]. We also score the tails: CRPS and the 0.05-quantile pinball loss on hours with negative prices, and CRPS and the 0.95-quantile pinball loss on hours above 300 AUD/MWh.

Statistical tests. Each model is trained with seeds 2026, 2027 and 2028. The main table averages separately computed seed scores; tail scores instead evaluate the seed-averaged quantiles. Differences are tested with the Diebold–Mariano test [14] on per-origin losses (MAE averaged over the 24 hours and the seeds; CRPS of seed-averaged quantiles), with a Newey–West long-run variance [32] with 48 lags because hourly origins overlap. Pooled tests use the three-region mean loss at each origin. For tail losses, each origin’s loss is its mean over the event hours in its horizon; pooling averages over regions with defined loss at that origin, and origins without events are dropped. Tail point estimates instead average all event-hour losses within each region before taking a three-region mean. Rolling-origin evaluation and recalibration follow the general recommendations of Tashman [33].

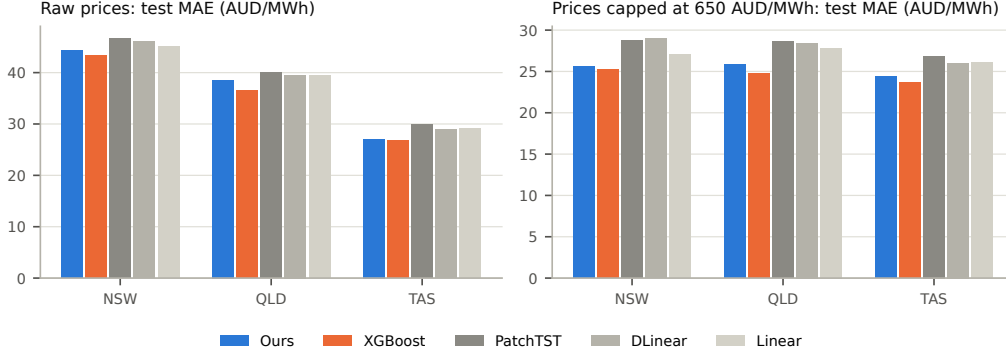


Figure 2: Test MAE per region in the main setting, raw prices (left) and prices capped at 650 AUD/MWh (right); means over three seeds.

286 *Implementation.* The models are implemented in PyTorch 2.5 and XGBoost
 287 3.2 and trained on NVIDIA L40 GPUs. Code and configurations reproduce
 288 every table and figure.

289 6. Results

290 6.1. Main comparison

291 Table 2 and Fig. 2 compare the models in the main setting, with quarterly
 292 refits and predispatch inputs, and Table 3 tests the differences. The dual-
 293 field model reduces the mean MAE of the same linear base from 37.60 to
 294 36.64 AUD/MWh on raw prices and from 26.72 to 25.29 on capped prices,
 295 and its CRPS from 26.04 to 25.07 and from 14.96 to 13.80 (all $p < 0.01$).
 296 It is more accurate than PatchTST, DLinear and the linear model on MAE
 297 under both price treatments and in every region ($p < 0.05$), and on capped
 298 CRPS ($p < 0.01$); on raw CRPS the differences to these three models are not
 299 significant. XGBoost has the lowest MAE and CRPS: our model is behind
 300 by 1.11 (raw) and 0.75 (capped) on MAE, most in QLD1 (2.10 raw); in TAS1
 301 the raw MAE difference is not significant.

302 6.2. When does the decomposition help?

303 Fig. 3 and Table 4 trace every model across three settings, and the first
 304 block of Table 5 isolates the dual fields against the same linear base in each.

305 On the static split (a single fit to 2015–2021), the fields *raise* MAE by 0.92
 306 (raw) and 0.95 (capped) ($p < 0.01$), and the dual-field model (41.54) is behind

Table 2: Main comparison on the 2023–2024 test set (17,521 hourly origins per region, 24-hour horizon). Every model is refit before each test quarter and uses the same inputs, including AEMO predispatch. Cells are means over three seeds; Mean MAE also shows the seed standard deviation. Ours includes the calibrator and the price floor; Linear base only is ours without them and without the dual fields. Lower is better; best in bold.

Model	MAE			Mean MAE	Window RMSE	90% AIS	CRPS
	NSW	QLD	TAS				
<i>Raw prices</i>							
Ours	44.27	38.57	27.07	36.64 ± 0.14	73.69	367.8	25.07
Linear base only	44.86	39.01	28.91	37.60 ± 0.03	74.35	401.2	26.04
XGBoost	43.34	36.47	26.76	35.52 ± 0.02	73.14	347.6	23.11
PatchTST	46.68	40.03	29.84	38.85 ± 0.07	75.94	355.3	24.33
DLinear	46.09	39.54	28.98	38.20 ± 0.09	76.11	338.7	23.62
Linear	45.16	39.51	29.10	37.92 ± 0.02	74.78	339.7	23.66
<i>Capped at 650</i>							
Ours	25.64	25.89	24.35	25.29 ± 0.17	36.07	165.8	13.80
Linear base only	26.77	27.46	25.92	26.72 ± 0.00	38.08	183.8	14.96
XGBoost	25.20	24.78	23.65	24.54 ± 0.02	35.51	160.7	13.10
PatchTST	28.77	28.67	26.87	28.10 ± 0.07	40.00	188.6	15.12
DLinear	29.01	28.44	26.02	27.82 ± 0.05	41.50	185.7	14.96
Linear	27.09	27.84	26.11	27.01 ± 0.01	38.42	186.1	14.98

the linear model with the same inputs (40.52). The fields lower the training loss but raise the error on the 2022 validation year: at the selected epoch in NSW1, validation MAE is 60.6 with the fields against 48.6 without. They fit the 2015–2021 level, which shifted in 2022 and again in 2023–2024 (Table 1). Normalizing each history window to remove the level did not help, because volatility shifted as well: the median window standard deviation of QLD1 prices is 2.2 times larger in the test period than in training (Appendix B).

The static-split result is metric-dependent: despite worsening MAE, the fields improve the same base’s CRPS (pooled differences -1.90 raw and -1.88 capped, $p < 0.01$, using seed-averaged quantiles). With quarterly refits, pooled MAE differences are not significant (-0.10 raw, -0.06 capped), while CRPS improves by about 1.1–1.3 ($p < 0.01$). Adding predispatch yields pooled gains in MAE (-0.34 raw, $p < 0.05$; -0.70 capped, $p < 0.01$) and CRPS (-1.13 , -0.94 , $p < 0.01$). These are not uniform regional gains: on raw prices, adding the fields changes MAE by -0.27 in NSW1 ($p = 0.334$), $+0.42$ in QLD1 ($p = 0.009$) and -1.16 in TAS1 ($p < 0.001$). Under capped

Table 3: Diebold–Mariano tests, ours minus comparator, in the main setting (negative favours ours). Δ columns pool the three regions; NSW, QLD and TAS give the raw-price MAE difference per region. MAE averages losses across seeds; CRPS scores seed-averaged quantiles, so its differences need not equal differences between the seed-mean scores in Table 2. Newey–West variance uses 48 lags. $*p < 0.05$, $**p < 0.01$.

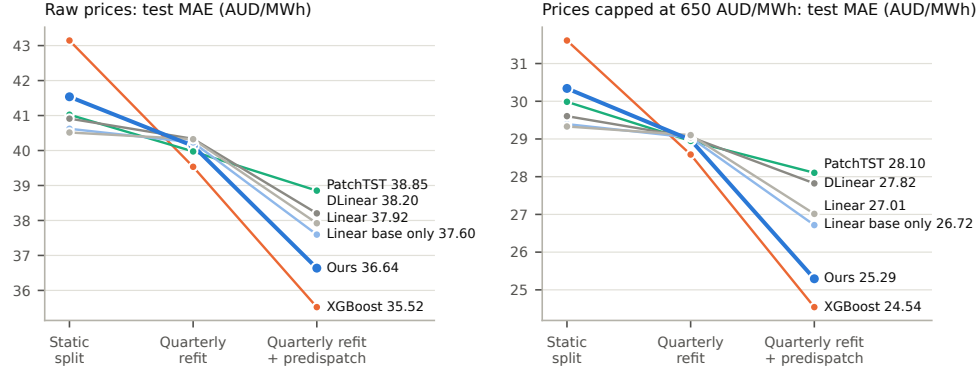
Comparator	Raw prices					Capped at 650	
	Δ MAE	NSW	QLD	TAS	Δ CRPS	Δ MAE	Δ CRPS
Linear base only	-0.96**	-0.59	-0.44	-1.84**	-1.42**	-1.42**	-1.18**
XGBoost	+1.11**	+0.93**	+2.10**	+0.31	+1.11**	+0.75**	+0.31**
PatchTST	-2.22**	-2.41**	-1.46**	-2.77**	-0.20	-2.81**	-1.69**
DLinear	-1.57**	-1.82*	-0.97*	-1.91**	+0.58	-2.53**	-1.46**
Linear	-1.28**	-0.89*	-0.93**	-2.03**	+0.53	-1.72**	-1.49**

prices it lowers MAE in all three regions ($p < 0.05$). Predispach lowers every model’s three-region mean MAE, and XGBoost’s most (39.53 to 35.52 raw), consistent with useful additional information from the operator’s dispatch simulation.

6.3. Diagnosis and corrections

To locate the remaining gap to XGBoost, we split the absolute-error difference by condition for the calibrated model before applying the lower bound. In QLD1, its seed-mean MAE gap is 2.9 AUD/MWh; for the seed-averaged point forecasts used in this diagnosis, the gap is 2.31 (38.56 against 36.25). Hours with a non-positive actual price (13% of hours) account for 47% of the latter gap, and hours in the lowest net-load quintile, largely the same midday solar hours, for 48%. The calibrated model forecast prices below -100 AUD/MWh in 0.23% of hours, against 0.01% of actual prices and none for XGBoost: the 3.2% of hours in which it forecast below -50 carry 31% of the gap. Linear heads extrapolate in asinh space, and the inverse transform can turn modest transformed-space deviations into deep negative prices. The fitted XGBoost comparator did not exhibit these extrapolations; this observation is not a general range-preservation guarantee for boosted trees.

Two corrections follow (second block of Table 5). Letting the model respond nonlinearly to predispach, by hinge features of the predispach price or by the calibrator, lowers MAE by 0.25–0.44 ($p < 0.01$ except the



Three-region means over seeds 2026-2028. Ours in the last setting includes the calibrator and the validation-chosen price floor. Axes do not start at zero.

Figure 3: Three-region mean test MAE of every model in three settings: a single fit on 2015–2021, quarterly refits, and quarterly refits with predispatch. Ours in the last setting includes the calibrator and the price floor. Means over three seeds; axes do not start at zero.

raw calibrator, $p < 0.05$); the calibrator is retained because it gives better negative-price tail scores than the hinge features. The validation-chosen price floor lowers MAE by a further 0.31 (raw) and 0.29 (capped), significantly in every region and most in QLD1 (-0.82 raw). In QLD1 the chosen point floors are -37 to -48 AUD/MWh in seven of the eight refits. Adding predispatch interconnector flows and limits does not help ($+0.04$ raw, $+0.19$ capped): the predispatch price already reflects the network constraints.

6.4. Price tails

Table 6 scores the tails. On negative-price hours, our final model has lower CRPS and 0.05-quantile pinball loss than all five comparators in that table under both price treatments, significantly lower in the pooled tests including against XGBoost (raw CRPS 14.52 against 15.96; $p < 0.001$). This does not establish superiority to every development variant: the calibrator without the lower bound achieves raw negative-price CRPS 14.17 and lower-tail pinball loss 4.30, compared with 14.52 and 4.74 after clipping. Capped negative-price CRPS likewise changes from 13.99 to 14.28. The lower bound therefore trades some tail accuracy for lower overall MAE. On capped prices the final model has the lowest spike-hour CRPS in the table (121.1 against 138.3 for XGBoost; $p < 0.01$); its difference to PatchTST is not significant, and PatchTST has slightly lower upper-tail pinball loss. On raw spike hours its CRPS difference

Table 4: Three-region mean test MAE across settings: static split, quarterly refit, quarterly refit with predispatch (PD). Means over three seeds; best per column in bold. Ours is the linear base with the dual fields; in the last setting it also includes the calibrator and the price floor.

Model	Raw prices			Capped at 650		
	Static	Refit	Refit + PD	Static	Refit	Refit + PD
Ours	41.54	40.13	36.64	30.34	28.98	25.29
Linear base only	40.62	40.23	37.60	29.39	29.04	26.72
XGBoost	43.15	39.53	35.52	31.61	28.59	24.54
PatchTST	41.02	39.97	38.85	29.99	28.95	28.10
DLinear	40.91	40.33	38.20	29.61	29.05	27.82
Linear	40.52	40.32	37.92	29.33	29.10	27.01

365 to XGBoost is not significant, while it is worse than PatchTST, DLinear and
366 the linear model ($p < 0.05$).

367 6.5. Case studies

368 Fig. 4 shows two test days chosen by fixed rules. On the QLD1 day
369 with the lowest mean price (1 October 2023), the linear base and the dual
370 fields keep the midday forecast positive; the calibrator brings it into negative
371 prices but overshoots to about -85 AUD/MWh, and the floor (-43) clips the
372 overshoot. The day’s MAE is 12.2 for our model, 14.1 for XGBoost and 17.3
373 for predispatch. On the NSW1 day on which predispatch most overstated
374 the peak (24 October 2024, predispatch 17,500 AUD/MWh against an actual
375 price below 300), every model is pulled up at 17:00; ours reaches about 470
376 and XGBoost about 280 (day MAE 57.1 and 52.8). Such phantom spikes
377 remain a failure mode.

378 6.6. Computational cost

379 Table 7 reports the size of each model and the wall-clock time of one
380 quarterly refit (NSW1, 2023Q3 test quarter, seed 2026) on one GPU. Every
381 model refits in four to six minutes; the dual-field model costs about 20% more
382 than its linear base and less than PatchTST. A full evaluation (eight refits,
383 three regions, three seeds) takes a few hours per model on one GPU.

Table 5: Ablations. Pooled Diebold–Mariano differences (negative is better). The first block compares the dual fields with the same linear base in each setting; each row of the second block is compared with the model it modifies. MAE averages seed losses; CRPS scores seed-averaged quantiles. These pooled tests do not imply improvement in every region. $^*p < 0.05$, $^{**}p < 0.01$.

Change	Raw prices		Capped at 650	
	Δ MAE	Δ CRPS	Δ MAE	Δ CRPS
<i>Fields minus linear base</i>				
Static split	+0.92**	-1.90**	+0.95**	-1.88**
Quarterly refit	-0.10	-1.28**	-0.06	-1.06**
Quarterly refit + predispatch	-0.34*	-1.13**	-0.70**	-0.94**
<i>Added to fields + predispatch</i>				
Hinge features on predispatch price	-0.25**	-0.95*	-0.39**	-0.26**
Calibrator	-0.31*	-0.26*	-0.44**	-0.22**
Price floor (after calibrator)	-0.31**	-0.03**	-0.29**	-0.01
Interconnector inputs (after floor)	+0.04	+0.04	+0.19**	-0.04

7. Discussion

Why decomposition has mixed value on a static split. The identity $c + e + r = x$ means the fields introduce no new observations, but their nonlinear representation can express relationships unavailable to a single linear map. In our static-split experiments this added structure worsens MAE while improving the same base’s CRPS. Lower training loss and higher validation MAE are consistent with sensitivity to the changing price regime; they do not isolate regime shift as the sole cause. Informer also performs poorly in the static screening (Appendix B), but differences in normalization alone do not establish the cause of that result.

Value of recalibration and predispatch. Refitting incorporates more recent observations, and predispatch supplies the operator’s forward-looking assessment of offers, demand and constraints. The pooled ablations support combining these inputs with the dual-field residual, but raw-price MAE gains from the fields are concentrated in TAS1 and do not extend to QLD1. Negative-price scores improve over the main-table baselines for the complete model; those gains cannot be attributed to the decomposition alone, because the calibrator and lower bound also change its forecasts.

Table 6: Probabilistic scores in the price tails, main setting, mean of the three regions, scoring seed-averaged quantiles. Lower is better for losses; best among the displayed models is bold. Coverage is evaluated over all test hours. $^{\dagger}/^{\ddagger}$: ours is significantly better/worse than that model in a pooled Diebold–Mariano test on per-origin event-hour losses ($p < 0.05$). Point estimates average event-hour losses within each region; tests first average within each origin, so their differences use a different weighting.

Model	Negative-price hours		Spike hours (>300)		
	CRPS	$q_{0.05}$ pinball	CRPS	$q_{0.95}$ pinball	90% coverage
<i>Raw prices</i>					
Ours	14.52	4.74	800.3	675.3	91.5%
Linear base only	16.21 †	5.82 †	882.3 †	777.2 †	93.1%
XGBoost	15.96 †	6.13 †	790.5	640.3	87.7%
PatchTST	17.99 †	7.51 †	745.2 ‡	600.3 ‡	90.0%
DLinear	17.25 †	7.33 †	740.3‡	591.7‡	89.1%
Linear	17.25 †	7.38 †	741.8 ‡	593.8 ‡	89.0%
<i>Capped at 650</i>					
Ours	14.28	4.83	121.1	77.2	90.5%
Linear base only	16.23 †	6.07 †	146.7 †	87.5 †	92.4%
XGBoost	15.47 †	6.13 †	138.3 †	91.0 †	86.2%
PatchTST	17.90 †	7.42 †	127.8	77.0	90.0%
DLinear	17.19 †	7.32 †	141.7 †	85.5	89.0%
Linear	17.31 †	7.44 †	141.9 †	85.5	88.7%

402 *The gap to XGBoost.* XGBoost uses these inputs more accurately on average,
403 and its fitted predictions avoid the extreme negative extrapolations observed
404 in the neural model. The calibrator and lower bound recover part of the
405 gap, but XGBoost remains ahead in the 100–300 AUD/MWh band and in
406 QLD1. For overall MAE on this benchmark, XGBoost is the stronger choice.
407 The dual-field model offers better negative-price scores than the main-table
408 baselines, while the unbounded calibrator variant offers still better negative-
409 price scores. Whether either improves storage or demand-response decisions
410 requires a separate decision-based evaluation.

411 *Limitations.* The study covers three regions of one market and a two-year
412 test period. The empirical lower bound is selected on validation data, but
413 the decision to introduce it followed a diagnosis of test-period errors; these

Table 7: Model size and wall-clock time of one quarterly refit (NSW1, 2023Q3, seed 2026), each run alone on an NVIDIA L40 shared with other users; times are indicative. XGBoost size is the total number of boosting rounds over the 24 per-hour models; Linear base only keeps the field modules, which are still fitted by the decomposition loss.

Model	Parameters	Seconds per refit
Ours	277,747	312
Linear base only	275,629	258
XGBoost	2,871 rounds	269
PatchTST	599,730	364
DLinear	78,120	252
Linear	83,232	237

414 results should therefore be confirmed on a later untouched period. On raw
415 spike hours the model is behind the linear baselines, and its regional gains
416 are heterogeneous. Predispatch run history is missing for October 2022, and
417 its horizon does not always cover all 24 target hours. The hard event-activity
418 penalty has no training gradient. Tail tests drop origins without events, so
419 their 48 lags count retained origins rather than fixed elapsed hours; reported
420 significance is conditional on that weighting and dependence approximation.
421 Hyperparameters were developed on the static split and were not re-tuned
422 for the recalibrated setting.

423 8. Conclusion

424 The value of trend–event decomposition in rolling electricity price fore-
425 casting depends on the metric, region and recalibration setting. On the static
426 split, adding the fields worsens the same linear base’s MAE but improves its
427 CRPS. With quarterly refits and predispatch inputs, pooled MAE and CRPS
428 improve, although raw-price MAE still worsens in QLD1. The complete
429 model, including the calibrator and empirical lower bound, improves MAE
430 over PatchTST, DLinear and a linear baseline and improves negative-price
431 scores over all five main-table comparators. The lower bound sacrifices some
432 negative-price accuracy relative to the unbounded calibrator. XGBoost retains
433 the lowest overall MAE and CRPS. Future work should test these tradeoffs
434 on an untouched period and in other markets, and model predispatch errors
435 explicitly to handle phantom spikes.

436 **CRediT authorship contribution statement**

437 **zl:** Conceptualization, Methodology, Software, Validation, Formal analysis,
438 Investigation, Data curation, Writing – original draft, Writing – review &
439 editing, Visualization.

440 **Declaration of competing interest**

441 The authors declare that they have no known competing financial interests
442 or personal relationships that could have appeared to influence the work
443 reported in this paper.

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447 **Data availability**

448 All data are public and were obtained from the Australian Energy Market
449 Operator [25, 26, 27]. Code and configurations that reproduce the results will
450 be made available.

451 **Appendix A. Field separation and static-split development**

452 In our forecasting transplant, the original Gabor DGF with globally learned
453 positions carries little of the historical spike signal, while the CTF absorbs it.
454 We quantify this by $R_e = \text{mean}_{\mathcal{S}} e^{(1)} / \text{mean}_{\mathcal{S}} x^{(1)}$, where $\mathcal{S}$ contains historical
455 positions with raw price above 300 AUD/MWh across all test windows. Both
456 numerator and denominator are in standardized asinh space, and repeated
457 positions in overlapping windows are retained. The ratio is -0.04 to 0.00 for
458 the Gabor DGF and 0.28 to 0.33 for detected events (Fig. A.5, Table A.8).
459 It is a transformed-signal diagnostic, not a fraction of monetary price or
460 explained variance. Fig. A.6 shows the static-split development, where most
461 MAE improvement came from target transformation and fundamental inputs.
462 Table A.9 reports the structural and detector ablations: suppression and the
463 median level matter, while event count and threshold have smaller effects.

Table A.8: Event-to-price mean ratio R_e on historical spike positions in test windows, measured in standardized asinh space.

Event field	NSW	QLD	TAS
Transplanted (Gabor)	-0.03	-0.00	-0.04
Detected events (ours)	0.33	0.33	0.28

Table A.9: Structural and detector ablations of the detected-event DGF on the static split, raw prices, three-region means over three seeds (MAE $\pm$ seed standard deviation). The last column gives the paired MAE difference to the detected-event DGF per seed; positive means the variant is worse.

Variant	MAE	Window RMSE	CRPS	Δ MAE vs detected (seeds)
<i>Structure</i>				
Transplanted dual field (Gabor DGF)	41.88 $\pm$ 0.22	79.21	28.43	+0.38 / +0.03 / +0.58
No decomposition (heads read raw history)	41.90 $\pm$ 0.11	79.31	28.40	+0.30 / +0.25 / +0.51
Adaptive attention atoms + echo	41.84 $\pm$ 0.05	79.01	28.09	+0.30 / +0.28 / +0.30
Detected events + daily echo	42.21 $\pm$ 0.11	79.93	28.22	+0.79 / +0.47 / +0.73
Detected-event DGF	41.54 $\pm$ 0.06	78.80	28.15	–
<i>Detector settings (8 events, 3-hour separation, learned threshold, median)</i>				
4 events	41.65 $\pm$ 0.03	78.85	28.25	+0.13 / +0.01 / +0.18
16 events	41.50 $\pm$ 0.08	78.76	28.15	-0.08 / -0.03 / -0.03
No suppression (1-hour separation)	41.85 $\pm$ 0.03	79.00	28.05	+0.36 / +0.21 / +0.35
6-hour separation	41.51 $\pm$ 0.05	78.77	28.28	-0.05 / -0.11 / +0.05
No threshold	41.63 $\pm$ 0.08	78.98	27.97	+0.21 / -0.03 / +0.07
Mean baseline	41.64 $\pm$ 0.03	78.91	28.02	+0.09 / +0.05 / +0.15

464 Appendix B. Designs that did not help

465 The following were tested with three seeds and are reported for complete-
466 ness (full records are in the code repository).

- 467 • *Window-normalized fields* on the linear base, with the residual rescaled
468 by the window’s price scale: raw MAE 44.29 against 41.54 unnormalized
469 (static split), because test-period volatility is about twice the training
470 volatility.
- 471 • *Fields on the quantiles only*, with asinh-space quantiles and the point
472 forecast from the linear base: the fields widened the 90% interval from
473 171 to 418 AUD/MWh and raised CRPS from 28.94 to 31.25 (static
474 split).
- 475 • *A future event field*, a three-component mixture for spike and trough

- 476 probabilities, did not beat a logistic regression on the same inputs in
477 precision–recall.
- 478 • *iTransformer and Informer* (static split, raw MAE 43.20 and 46.65) were
479 less accurate than PatchTST (41.02) and DLinear (40.91); Informer’s
480 validation loss was lowest after one to five epochs.
 - 481 • *Predispatch interconnector inputs*: no gain (Section 6.3).

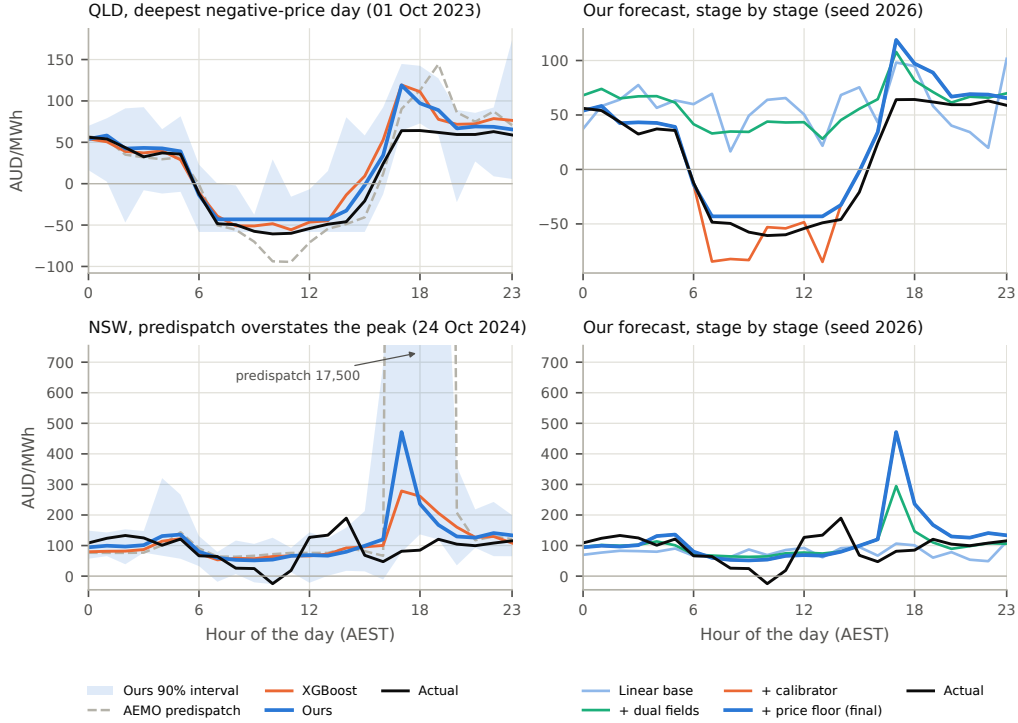


Figure 4: Case studies, chosen by fixed rules on the test split. Top: QLD1 day with the lowest mean price. Bottom: NSW1 day on which predispatch most overstated the peak (forecast above 1,000, actual below 300 AUD/MWh). Left: our forecast with its 90% interval, XGBoost, predispatch and the actual price. Right: cumulative component outputs of the same jointly trained seed-2026 model, not independently retrained ablations. On the NSW1 day the lower bound is not binding, so the calibrated and final forecasts coincide.

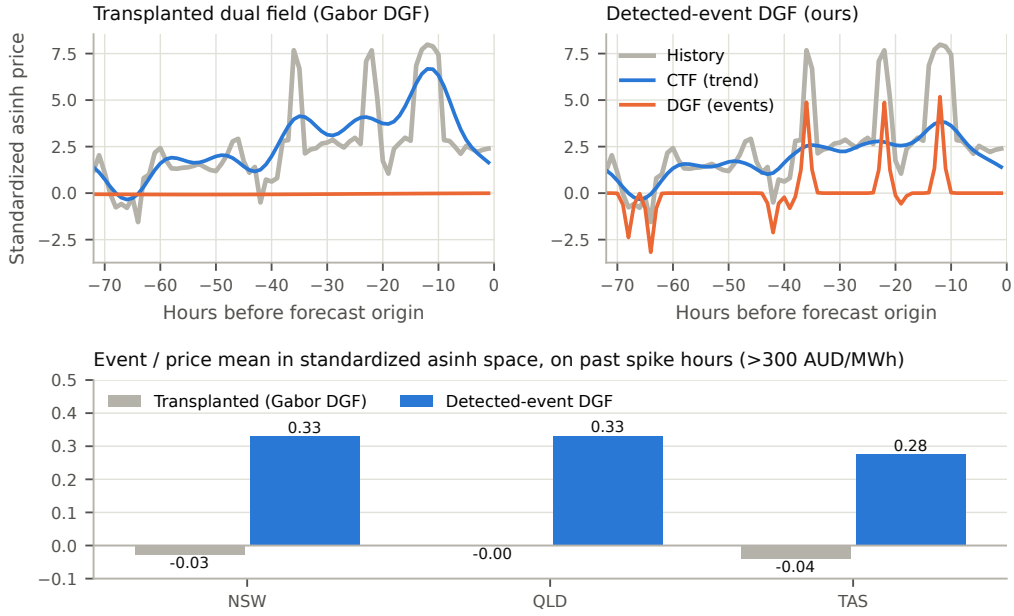


Figure A.5: Field separation. Top: one NSW1 test window in standardized asinh price; the Gabor DGF stays near zero (left), whereas detected events capture localized deviations (right). Bottom: event-to-price mean ratio R_e in standardized asinh space, restricted to historical raw-price spikes above 300 AUD/MWh. This is not a monetary contribution share.

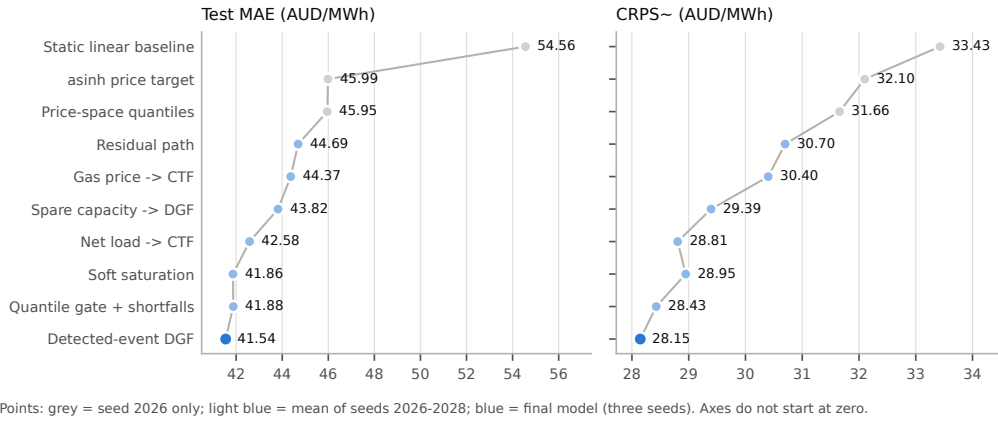


Figure A.6: Development on the static split: three-region mean test MAE and CRPS on raw prices; each step changes one component. Grey: seed 2026 only; blue: means over seeds 2026–2028. Axes do not start at zero.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used [tool name] in order to [purpose]. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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